

Assignment 1

Data Lakehouse With Snowflake

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1. Introduction

The digital economy thrives on data-driven insights. Platforms like YouTube generate massive amounts of content daily, offering opportunities to understand audience preferences and global trends. This project builds a Data Lakehouse in Snowflake to process, clean, and analyze YouTube Trending Videos from 2020 to 2024 across ten countries (India, USA, UK, Germany, Canada, France, Brazil, Mexico, South Korea, and Japan).

The goal was to ingest, transform, and consolidate multi-source data (CSV and JSON files) into a structured, query-ready dataset, enabling analysis of trending categories, user engagement, and content strategies. The dataset includes metadata such as views, likes, comments, category information, and daily trending videos, providing a rich foundation for insights across multiple regions.

2. Project Overview

The project was executed in **four main parts**:

- **Data Ingestion** – Load data into Snowflake via Azure Blob Storage and create consolidated internal tables.
- **Data Cleaning** – Handle duplicates, missing values, and invalid entries to ensure dataset integrity.
- **Data Analysis** – Perform SQL-driven analysis on top videos, trending categories, engagement, and channel performance.
- **Business Question** – Recommend a content strategy for a new YouTube channel based on global and regional trends.

3. Part 1: Data Ingestion

The first step was building a robust data pipeline in Snowflake.

3.1 Database and Stage Setup

A dedicated database (**assignment_1**) was created to isolate resources. A stage (**stage_assignment**) connected Snowflake securely to Azure Blob Storage using a SAS token. Files in both CSV (trending data) and JSON (category data) formats were verified for accessibility.

3.2 External Tables

External tables allowed querying raw data without moving it into managed storage.

- **ex_table_youtube_trending** mapped CSV fields like video_id, title, publishedAt, view_count, likes, dislikes, and comment_count.

	{?} VALUE	VIDEO_ID	TITLE	PUBLISHEDAT	CHANNELID	CHANNELTITLE	CATEGORYID	TRENDING_DATE	#
1	{ "c1": "7rlwxSPUcQk", "c10": "4198", "c11": "15777", "c2": "7rlwxSPUcQk"	7rlwxSPUcQk	ON EST POSITIF AU COVID-19 (coronavirus)	2020-08-11	UCpWaR3gNAQGsX48cllQC0qw	Tibo InShape	17	2020-08-12	
2	{ "c1": "AcBd_RH9JSw", "c10": "2767", "c11": "8030", "c2": "AcBd_RH9JSw"	AcBd_RH9JSw	PASSER UNE NUIT DANS LA PISCINE DE LA VI	2020-08-11	UCUj7mwOyySFzZUkq4H29nug	LeBouseuh	24	2020-08-12	
3	{ "c1": "JvM8P6kKgD0", "c10": "187", "c11": "1148", "c2": "JvM8P6kKgD0"	JvM8P6kKgD0	FRANGLISH - My Salsa feat. Tory Lanez (clip o	2020-08-12	UCnwf00-LGEg8y9bEQTaSgQ	Franglish	10	2020-08-12	
4	{ "c1": "JfgeD7xWy-w", "c10": "418", "c11": "3213", "c2": "JfgeD7xWy-w"	JfgeD7xWy-w	L'ÉTÉ LE PLUS ECLATÉ DE MA LIFE	2020-08-11	UCMj2VQ3-8zpyeAl7dU0T-Bg	Fahd El	23	2020-08-12	
5	{ "c1": "P3IkBxra3a8", "c10": "90", "c11": "384", "c2": "JE P3IkBxra3a8"	P3IkBxra3a8	JE L'ESSAYE ENFIN ! DUCATI HYPERMOTARD	2020-08-12	UC-uy7_1hColMpQ_2rs-kY6w	KIKANINAC	24	2020-08-12	

- **ex_table_youtube_category** extracted category metadata from JSON files using LATERAL FLATTEN to map categoryId and title.

	CATEGORYID	ETAG	KIND	ASSIGNABLE	CHANNEL_ID	CATEGORY_TITLE
1	1	IfWa37JGcqZs-jZeAyFGkbeh8bc	youtube#videoCategory	TRUE	UCBR8-60-B28hp2BmDPdntcQ	Film & Animation
2	2	5XGyls7zkjHh5940dsT5862m1Y	youtube#videoCategory	TRUE	UCBR8-60-B28hp2BmDPdntcQ	Autos & Vehicles
3	10	HCjFMARbBeWjpm6PDfReCOMOZGA	youtube#videoCategory	TRUE	UCBR8-60-B28hp2BmDPdntcQ	Music
4	15	ra8H7xyAfmE2FewsDabE3TUSq10	youtube#videoCategory	TRUE	UCBR8-60-B28hp2BmDPdntcQ	Pets & Animals
5	17	7mqChSjogdF3hSIL-88BfDE-W8M	youtube#videoCategory	TRUE	UCBR8-60-B28hp2BmDPdntcQ	Sports

3.3 Structured Internal Tables

For better query performance and transformations, external tables were materialized into internal tables:

- **table_youtube_trending** included country codes derived from file names for regional analysis.

	<u>A VIDEO_ID</u>	<u>A TITLE</u>	<u>PUBLISHEDAT</u>	<u>A CHANNELID</u>	<u>A CHANNELTITLE</u>	# CATEGORYID	<u>TRENDING_DATE</u>	# VIEW_COUNT	# LIKES	# DISLIKES	# COMMENT_COUNT	<u>A COUNTRY</u>
1	J78aPJ3VyNs	I left youtube for a month and THIS is what happened.	2020-08-11	UCYzPXprv5Y-Sf0g4vX-m6g	jacksepticeye	24	2020-08-12	2038853	353790	2628	40228	GB
2	9nidKH8cM38	TAXI CAB SLAYER KILLS 'TO KNOW HOW IT FEELS'	2020-08-11	UCFmbX7fHWZfuWdJAML0babA	Eleanor Neale	27	2020-08-12	236830	16423	209	1642	GB
3	M9Pmf9AB4Mo	Apex Legends Stories from the Outlands - "The Endo	2020-08-11	UC0ZV6M2THA81QT9hrVWJG3	Apex Legends	20	2020-08-12	2381688	146739	2794	16549	GB

- **table_youtube_category** mapped categoryId to category_title per country.

	<u>A COUNTRY</u>	<u>A CATEGORYID</u>	<u>A CATEGORY_TITLE</u>
1	GB	1	Film & Animation
2	GB	2	Autos & Vehicles
3	GB	10	Music

3.4 Final Consolidated Table

- **table_youtube_final** joined trending videos with category information using **LEFT JOIN**. Each record received a unique ID via **UUID_STRING()**. All metrics and country context were preserved, resulting in **2,667,041 records** for cleaning and analysis.

	<u>A ID</u>	<u>A VIDEO_ID</u>	<u>A TITLE</u>	<u>PUBLISHEDAT</u>	<u>A CHANNELID</u>	<u>A CHANNELTITLE</u>	# CATEGORYID	<u>A CATEGORY_TITLE</u>	<u>TRENDING_DATE</u>	Query Details	***
1	c57028ee-7cbb-4d4c-aa32-c770e0b56b	ENv7p2BVAEo	Thankyou For Your Support Uma Telugu Traveller	2022-07-23	UC-elz6w4sbn_5LB37pJTRA	Uma Telugu Traveller	19	Travel & Events	2022-0	Query duration	4.2s
2	f1b46152-b0dc-4396-9b4c-022d84c8fe	iKxIRL_YabA	Cute love story Part 3 #shorts #pidivya	2022-07-22	UCMbXauD-GmTmkKE_OiIFuQ	Pidivya Official	22	People & Blogs	2022-0	Rows	2,667,041
3	61727688-c470-4fa4-91b8-3c01f2c7f14	41RILqnwyJ0	Hahaha😂	2022-07-21	UCipVQOsmadQ-vDHYoPIp38A	DURGARAO CHANNEL OI	22	People & Blogs	2022-0	Query ID	01be83d4-0002-46bc-...

4. Part 2: Data Cleaning

Data cleaning ensured the dataset was consistent, accurate, and ready for analysis.

Q1: Duplicate category titles

Some **category_title** appeared across multiple categoryIds. **One** duplicate title was identified.

	<u>A</u> CATEGORY_TITLE
1	Comedy

Purpose: Avoid misclassifying videos in analysis.

Q2: Country-specific categories

One **category_title** appeared in only a single country, highlighting regional classification differences.

	<u>A</u> CATEGORY_TITLE
1	Nonprofits & Activism

Purpose: Understanding country-specific categories prevents misinterpretation when comparing trends.

Q3: Missing category_title in table_youtube_final

Some records had a valid categoryId but **NULL** category_title (e.g., categoryId=29).

	# CATEGORYID	:
1		29

Purpose: Identifying missing titles ensures completeness in aggregations.

Q4: Filling missing category titles

NULL category_titles were updated using **table_youtube_category**, filling 1,563 rows. The **MAX()** function was used to ensure only one title was assigned per **categoryId**.

	# number of rows updated
1	1563

Purpose: Ensures all videos have complete category information for analysis.

Q5: Videos without channel title

Only **one** video lacked a channelTitle.

	A TITLE
1	Kala Official Teaser Tovino Thomas Rohith V S Juvis Productions Adventure Company

Purpose: Allows filtering or handling incomplete data in channel-level analysis.

Q6: Removing invalid records

Records with **video_id= '#NAME?'** were deleted, removing 32,081 invalid rows.

	# number of rows deleted
1	32081

Purpose: Ensures the dataset contains only valid, real video records.

Q7: Identifying duplicates

Duplicates were defined as multiple rows having the same **video_id**, **country**, and **trending_date** but differing metrics (e.g., **view_count**, **likes**). Using the **ROW_NUMBER()** function:

- Ranked duplicates by **view_count** descending.
- Retained the highest view count per group.
- Stored extra duplicates in **table_youtube_duplicates**.

	A status
1	Table TABLE_YOUTUBE_DUPLICATES successfully created.

Purpose: Isolates inconsistent records while retaining the most relevant data.

Q8: Removing duplicates

All rows identified in **table_youtube_duplicates** were deleted from **table_youtube_final**.

	# number of rows deleted
1	37466

Q9: Final row count

After all cleaning steps, the row count of **table_youtube_final** was **2,597,494**, confirming successful removal of duplicates and data integrity for analysis.

	# FINAL_ROW_COUNT
1	2597494

5. Part 3: Data Analysis

After completing the data cleaning stage, we conducted a detailed **analysis of the YouTube trending videos dataset**. The aim of this section is to derive insights into **trending content, popular categories, viral videos, and channel performance** across countries and time periods. The analysis was performed using **Snowflake SQL queries**, which are detailed below.

5.1 Top 3 Videos by Views per Country on 2024-04-01 (Gaming Category)

Using a **common table expression (CTE)** with **ROW_NUMBER()**, we partitioned the data by **country** and ranked videos in descending order of **view_count**. Only the top 3 videos per country were selected.

SQL Logic:

- Filter **table_youtube_final** for **trending_date = '2024-04-01'** and **category_title = 'Gaming'**.
- Assign a ranking number to videos per country using **ROW_NUMBER()** based on **view count**.
- Select only the top 3 videos per country.

	COUNTRY	TITLE	CHANNELTITLE	VIEW_COUNT	RK	Query Details
1	BR	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1	Query duration 2.3s
2	BR	IShowSpeed x MC Kevin O Chris - Amar de (Official Music Video)	IShowSpeed	2971782	2	Rows 30
3	BR	Confrontation - The Skibidi Saga 05	Maxedy	2323375	3	Query ID 01be8413-0002-46bc-0...
4	CA	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1	Show more
5	CA	If my viewers break my secret rule, I ban them	DougDoug	2988844	2	COUNTRY
6	CA	Confrontation - The Skibidi Saga 05	Maxedy	2323375	3	BR 3
7	DE	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1	CA 3
8	DE	If my viewers break my secret rule, I ban them	DougDoug	2988844	2	DE 3
9	DE	Season 3 Warzone Launch Trailer - Rebirth Island Call of Duty: Warzone	Call of Duty	2311131	3	
10	FR	DAGGER DUCHESS - New Tower Troop! (Official Music Video)	Clash Royale	4923026	1	

Insights:

- This query highlighted the **most viewed gaming content across countries**, revealing regional differences in gaming preferences.

- The output can help content creators target specific regions with highly engaging gaming videos.

5.2 Number of Videos Mentioning “BTS” per Country

Next, we investigated videos containing “**BTS**” in their titles, reflecting the popularity of the band in different regions. The query counted **distinct video_id per country** where the title contained “BTS” (case-insensitive). Results were sorted in descending order of count.

SQL Logic:

- Use **LOWER(title) LIKE '%bts%'** to match titles containing BTS.
- Group by **country** and count distinct videos.
- Order count by **Desc** to find the countries with the highest number of BTS-related videos.

	A COUNT :	# CT
1	KR	468
2	IN	288
3	US	268
4	CA	262
5	MX	254
6	JP	251
7	DE	242
8	GB	223
9	BR	186
10	FR	167

Insights:

- This query provided a **measure of BTS’s global influence**, identifying countries where videos mentioning BTS were trending frequently.
- Countries with higher counts could indicate stronger fan engagement or cultural resonance.
- After **Korea** , **India** and **the US** have more fans for BTS.

5.3 Monthly Most Viewed Video per Country (2024) and Likes Ratio

The third analysis focused on **monthly trends in 2024**, identifying the **most viewed video per country for each month** and calculating a likes-to-views ratio (**likes_ratio**) to evaluate audience engagement.

SQL Logic:

- Extract the year and month using **DATE_TRUNC('MONTH', trending_date)**.
- Partition by **country** and **month**, and rank videos by **view_count** using **ROW_NUMBER()**.
- Select only the top-ranked video for each month and country.
- Calculate **likes_ratio = (likes / view_count) * 100**, rounded to 2 decimals.

	COUNTRY	YEAR_MONTH	TITLE	CHANNELTITLE	CATEGORY_TITLE	VIEW_COUNT	LIKES_RATIO	Query Details
1	BR	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	139504939	3.20	Query duration 1.7s
2	CA	2024-01-01	Still Here Season 2024 Cinematic - League of Legend	League of Legends	Gaming	104159411	1.69	Rows 40
3	DE	2024-01-01	Still Here Season 2024 Cinematic - League of Legend	League of Legends	Gaming	104159411	1.69	Query ID 01be841f-0002-4650-0...
4	FR	2024-01-01	Still Here Season 2024 Cinematic - League of Legend	League of Legends	Gaming	104159411	1.69	Show more
5	GB	2024-01-01	Still Here Season 2024 Cinematic - League of Legend	League of Legends	Gaming	104159411	1.69	COUNTRY
6	IN	2024-01-01	Protect \$500,000 Keep It!	MrBeast	Entertainment	85458562	4.21	BR 4
7	JP	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	137639799	3.22	CA 4
8	KR	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	143955997	3.16	DE 4
9	MX	2024-01-01	Survive 100 Days Trapped, Win \$500,000	MrBeast	Entertainment	137639799	3.22	
10	US	2024-01-01	Grand Theft Auto VI Trailer 1	Rockstar Games	Gaming	166323421	6.73	

Insights:

- The query revealed **monthly performance trends**, highlighting which videos captured the largest audience per country and most of the videos are from MrBeast channel.
- The likes ratio provided an **engagement metric**, helping to distinguish between merely popular videos and those that resonated strongly with viewers.

5.4 Most Popular Category per Country (Since 2022)

This analysis aimed to identify **which content categories dominate the YouTube trending landscape** in each country since 2022. The process involved counting **distinct videos per category** per country, calculating each category's **percentage of total videos**, and ranking categories using **ROW_NUMBER()**.

SQL Logic:

- Count distinct videos per **category_title** and **country**.
- Calculate total distinct videos per country.
- Compute percentage share of each category using $(\text{category_videos} / \text{total_country_videos}) * 100$.
- Select the top-ranked category per country.

	A COUNTRY	A CATEGORY_TITLE	# CATEGORY_VIDEOS	# TOTAL_COUNTRY_VIDEOS	A PERCENTAGE
1	BR	Entertainment	5417	23760	22.80
2	DE	Entertainment	7709	30719	25.10
3	FR	Entertainment	7548	32849	22.98
4	GB	Entertainment	5643	27855	20.26
5	IN	Entertainment	21281	50250	42.35
6	JP	Entertainment	5658	17627	32.10
7	KR	Entertainment	5122	15175	33.75
8	MX	Entertainment	4195	17532	23.93
9	CA	Gaming	6594	30869	21.36
10	US	Gaming	6226	28799	21.62

Insights:

- This query highlighted **Entertainment** is the dominant category across 8 countries.
- The results are valuable for new creators and marketers to identify **high-demand categories** that have the potential to trend across different countries.

5.5 Channel with Most Distinct Videos

Finally, we analyzed **channel performance**, identifying the **channel that produced the highest number of unique videos** in the dataset. This metric reflects productivity and potential influence on trending content.

SQL Logic:

- Count distinct **video_id** per **channeltitle**.

- Sort in descending order and select the top channel.

	A CHANNELTITLE	# DISTINCT_VIDEOS
1	Vijay Television	2049

Insights:

The result identified Vijay Television as the **most prolific channel** contributing to trending content with 2049 videos.

6. Part 4: Business Question

If you were to launch a new YouTube channel tomorrow, which category (excluding “Music” and “Entertainment”) should you focus on to maximize chances of trending globally? Will this strategy work consistently across all countries?

6.1 Step 1: Identify Top-Performing Categories Globally

The first step was to **evaluate category performance at a global level**, excluding the already dominant “Music” and “Entertainment” categories. The analysis considered:

- Total **distinct trending videos** per category, which shows the volume of content reaching trending status.
- **Average views** per category, reflecting audience engagement and popularity.

SQL Logic:

- Filter **table_youtube_final** to exclude **Music** and **Entertainment**.
- Group by **category_title** to count unique **video_id** and calculate **AVG(view_count)**.
- Order the results by the number of distinct videos and average views in **descending** order to identify the strongest candidates.



	A CATEGORY_TITLE	# DISTINCT_VIDEOS	# AVG_VIEWS
1	Sports	43323	1199500
2	People & Blogs	42876	1400017
3	Gaming	41615	1321615
4	Comedy	18605	1491988
5	News & Politics	13026	1100605
6	Howto & Style	11139	861314
7	Film & Animation	8189	2441205
8	Science & Technology	8188	2189231
9	Education	7600	1226924
10	Autos & Vehicles	7517	654593
11	Travel & Events	2991	780852
12	Pets & Animals	1803	946473
13	Nonprofits & Activism	212	1267903

Insights:

- This query revealed categories that consistently generate trending content globally.
- Categories with high numbers of trending videos and high average views are strong candidates for a **new channel strategy**.
- For example, categories like **Sports, People & blogs, Gaming** often appear as high-performing globally.

6.2 Step 2: Identify Top 3 Categories per Country

Next, we evaluated **regional preferences** by ranking categories within each country. This helps determine whether a single global category strategy can work across all regions or if a **localized approach** is needed.

SQL Logic:

- For each **country**, count distinct videos per **category_title**.
- Use **ROW_NUMBER()** to rank categories per country and Select the top 3 categories.

	A COUNTRY	A CATEGORY_TITLE	# DISTINCT_VIDEOS
1	BR	Sports	7122
2	BR	Gaming	5865
3	BR	People & Blogs	4536
4	CA	Gaming	9913
5	CA	Sports	6878
6	CA	People & Blogs	4343
7	DE	Sports	7567
8	DE	Gaming	6386
9	DE	People & Blogs	5604
10	FR	Gaming	7788
11	FR	Sports	6828
12	FR	Comedy	4116
13	GB	Sports	9590
14	GB	Gaming	7783
15	GB	People & Blogs	4172

Insights:

- The results highlighted that **top-performing categories differ across countries**.
- Some countries may favor **Gaming**, while others trend towards **Sports** or **People & Blogs** content.
- This shows that while certain categories perform well globally, **a one-size-fits-all approach is not guaranteed to succeed in every country**.

6.3 Step 3: Aggregate Top 3 Rankings Across All Countries

To determine **consistency across regions**, we aggregated the top 3 category rankings for all countries:

- Counted how many countries ranked each category as **#1, #2, or #3** in their top 3.

- Calculated the **total number of countries** where the category appeared in the top 3.

SQL Logic:

- Use a CTE (**top3**) to compute rankings per country.
- Aggregate top1, top2, top3 counts across countries.
- Sort categories by their top1 count, then top2 and top3 counts.

	A CATEGORY_TITLE	# TOP1_COUNT	# TOP2_COUNT	# TOP3_COUNT	# TOTAL_COUNTRIES
1	Gaming	5	3	1	9
2	Sports	3	4	2	9
3	People & Blogs	2	2	5	9
4	Comedy	0	1	2	3

Insights:

- Categories with high **top1_count** across multiple countries are **strong candidates for a global channel strategy**.
- Categories with highly variable rankings across countries indicate the need for **regional content adaptation**.
- This step provides a **data-driven approach** to deciding whether to launch a globally focused channel or to adopt a **country-specific content strategy**.

6.4 Recommendations

Based on the global and regional analysis, the **Gaming**, **Sports**, and **People & Blogs** are the strongest options for starting a YouTube channel today. These categories consistently perform well across nine countries, highlighting their wide appeal and potential for growth.

From a personal perspective, I would choose the **Gaming** category as my preferred area to start a channel. Gaming is not only performing strongly worldwide but also aligns well with my personal interests and skills—I enjoy playing games and believe I can create engaging content in this space.

Although **Sports** is another highly successful category, it presents significant challenges for individual creators. Sports broadcasting rights are usually owned by large organizations such as



ESPN, Star Sports, and Sony. As a result, only licensed organizations can legally stream and monetize such content, making it impractical for an independent YouTube creator.

The **People & Blogs** category is also a strong candidate globally; however, I would personally avoid it. This category often requires sharing personal lifestyle content or conducting product reviews, which may involve additional expenses for purchasing items. Since I do not prefer sharing too much about my personal life or investing in products solely for review purposes, I would not pursue this path.

Final Recommendation:

Gaming offers the best combination of popularity, creativity, and accessibility without legal or financial barriers.

A single global strategy is unlikely to succeed everywhere, even within the same category. Gaming trends vary by region—some countries prefer PUBG, others Clash of Clans, Clash Royale, or Chess. My approach would focus on countries with similar gaming preferences and tailor content accordingly, ensuring higher engagement and adaptability.

7. Conclusion

This project analyzed YouTube categories across multiple countries, uncovering both global and regional trends. The results showed that categories such as **Gaming, Sports, and People & Blogs** are among the strongest performers across nine countries, while some categories remain unique to specific regions. This shows the importance of considering each country's trends rather than assuming universal success.

Overall, the study highlights that YouTube success depends on balancing **universal appeal with localized strategies**. While gaming stands out as a strong global opportunity, creators must adapt to the unique preferences of each region, for example, focusing on different games that trend in different countries. A regionally adaptive, data-driven approach is more effective than a one-size-fits-all strategy.

8. References

Snowflake Inc. (2024). *Snowflake Documentation*. Retrieved from <https://docs.snowflake.com>

Microsoft Azure. (2024). *Azure Blob Storage Overview*. Retrieved from <https://learn.microsoft.com/en-us/azure/storage/blobs/>

William So (2025, August) Tutorial 1 & 2. Data Warehousing & Data Lake. Big Data Engineering. UTS TD School.

9. Appendices

Since I couldn't include all the setup screenshots (Part 1) in the main report due to the word limit, I have attached them in the appendices.

1. Creating a Database

	A status
1	ASSIGNMENT_1 already exists, statement succeeded.

2. Creating Stage

	A status
1	Stage area STAGE_ASSIGNMENT successfully created.

3. LIST of stage_assignment

	A name	:	# size	A md5	A last_modified
1	azure://bigdataeng.blob.core.windows.net/assignment-1/youtube-ca		10096	e5f8d4189665ae35616bd65ff261c42b	Sun, 17 Aug 2025 06:19:55 GMT
2	azure://bigdataeng.blob.core.windows.net/assignment-1/youtube-ca		10096	e5f8d4189665ae35616bd65ff261c42b	Sun, 17 Aug 2025 06:19:55 GMT
3	azure://bigdataeng.blob.core.windows.net/assignment-1/youtube-ca		10096	e5f8d4189665ae35616bd65ff261c42b	Sun, 17 Aug 2025 06:19:55 GMT
4	azure://bigdataeng.blob.core.windows.net/assignment-1/youtube-ca		10096	e5f8d4189665ae35616bd65ff261c42b	Sun, 17 Aug 2025 06:19:55 GMT
5	azure://bigdataeng.blob.core.windows.net/assignment-1/youtube-ca		10096	e5f8d4189665ae35616bd65ff261c42b	Sun, 17 Aug 2025 06:19:55 GMT

4. Creation of ex_table_youtube_trending

	A status
1	Table EX_TABLE_YOUTUBE_TRENDING successfully created.

5. Preview of ex_table_youtube_trending

	{?} VALUE
1	{ "c1": "KJl2qg5F-9E", "c10": "970", "c11": "3311", "c2": "Bonz MC - HOLLYWOOD (Snippet)", "c3": "2020-08-11T18:00:03Z", "c4": "UCGh8tmH9x9njal2mXfh2fyg", "c5": "CrhymeTV", "
2	{ "c1": "K0vYnOn7wZl", "c10": "435", "c11": "866", "c2": "Nik hat heftige Probleme in Köln! 🤔 #1925 Köln 50667", "c3": "2020-08-11T17:00:09Z", "c4": "UCnrvUg5MJWPDSrv_voT7Aq
3	{ "c1": "2bbn9b79LRc", "c10": "144", "c11": "364", "c2": "Camper Tour 2020 - ROADTRIP durch Österreich Episode #2 AnaJohnson", "c3": "2020-08-11T18:00:00Z", "c4": "UCBt8RY61t
4	{ "c1": "Zv-3qNnAMaM", "c10": "124", "c11": "229", "c2": "Ich TESTE SHEIN BIKINIS (try on haul) - UNMÖGLICH *wtf i'm shook* BYE", "c3": "2020-08-12T09:00:08Z", "c4": "UCccDoH6Qp
5	{ "c1": "7clgQLneouU", "c10": "375", "c11": "1051", "c2": "STATEMENT zu meinem BANI 🍷 Bald vielleicht PERMABAN...? Papaplatte Highlights", "c3": "2020-08-11T12:00:02Z", "c4": "U

6. Preview of Structured ex_table_youtube_trending

	{?} VALUE	VIDEO_ID	TITLE	PUBLISHEDAT	CHANNELID	CHANNELTITLE	CATEGORYID	TRENDING_DATE	VIEW_COUNT	LIKES	DISLIKES	COMMENT_COUNT
1	{ "c1": "J78aPJ3VyNs",	J78aPJ3VyNs	I left youtube for a mc	2020-08-11	UCYzPXprvI5Y-Sf0z	jacksepticeye	24	2020-08-12	2038853	353790	2628	40228
2	{ "c1": "9nidKH8cM38"	9nidKH8cM38	TAXI CAB SLAYER KIL	2020-08-11	UCFmbX7frWZfuWc	Eleanor Neale	27	2020-08-12	236830	16423	209	1642
3	{ "c1": "M9Pmf9AB4Mc	M9Pmf9AB4Mc	Apex Legends Stor	2020-08-11	UC0ZV6M2THA81C	Apex Legends	20	2020-08-12	2381688	146739	2794	16549
4	{ "c1": "kgUV1MaD_M8	kgUV1MaD_M8	Nines - Clout (Offic	2020-08-10	UCvDkzrj8ZPIBqRdt	Nines	24	2020-08-12	613785	37567	669	2101
5	{ "c1": "4926Mv4_WCA	4926Mv4_WCA	I don't know what im	2020-08-11	UCTinbF-Q-fVthA0q	CaseyNeistat	22	2020-08-12	940036	87113	1860	7052

7. Creation of ex_table_youtube_category

	status
1	Table EX_TABLE_YOUTUBE_CATEGORY successfully created.

8. Preview of ex_table_youtube_category

	{?} VALUE
1	{ "etag": "kBcCr3i9kLHHU79W4lp5196LDptl", "items": [{ "etag": "lFWa37JGcqZs-jZeAyFGkbeh6bc", "id": "1", "kind": "youtube#videoCategory", "snippet": { "assignable": true, "channelId": "UCBR8-60-B28hp2BmDPdntcQ
2	{ "etag": "kBcCr3i9kLHHU79W4lp5196LDptl", "items": [{ "etag": "lFWa37JGcqZs-jZeAyFGkbeh6bc", "id": "1", "kind": "youtube#videoCategory", "snippet": { "assignable": true, "channelId": "UCBR8-60-B28hp2BmDPdntcQ
3	{ "etag": "kBcCr3i9kLHHU79W4lp5196LDptl", "items": [{ "etag": "lFWa37JGcqZs-jZeAyFGkbeh6bc", "id": "1", "kind": "youtube#videoCategory", "snippet": { "assignable": true, "channelId": "UCBR8-60-B28hp2BmDPdntcQ
4	{ "etag": "kBcCr3i9kLHHU79W4lp5196LDptl", "items": [{ "etag": "lFWa37JGcqZs-jZeAyFGkbeh6bc", "id": "1", "kind": "youtube#videoCategory", "snippet": { "assignable": true, "channelId": "UCBR8-60-B28hp2BmDPdntcQ

9. Preview of Structured ex_table_youtube_category

	CATEGORYID	ETAG	KIND	ASSIGNABLE	CHANNEL_ID	CATEGORY_TITLE
1	1	lFWa37JGcqZs-jZeAyFGkbeh6bc	youtube#videoCategory	TRUE	UCBR8-60-B28hp2BmDPdntcQ	Film & Animation
2	2	5XGylls7zkjHh5940dsT5862m1Y	youtube#videoCategory	TRUE	UCBR8-60-B28hp2BmDPdntcQ	Autos & Vehicles
3	10	HCjFMARbBeWjpm6PDfReCOMOZGA	youtube#videoCategory	TRUE	UCBR8-60-B28hp2BmDPdntcQ	Music
4	15	ra8H7xyAfmE2FewsDabE3TUSq10	youtube#videoCategory	TRUE	UCBR8-60-B28hp2BmDPdntcQ	Pets & Animals

10. Creation of table_youtube_trending

	status
1	Table TABLE_YOUTUBE_TRENDING successfully created.



11. Preview of table_youtube_trending

	VIDEO_ID	TITLE	PUBLISHEDAT	CHANNELID	CHANNELTITLE	CATEGORYID	TRENDING_DATE	VIEW_COUNT	LIKES	DISLIKES	COMMENT_COUNT	COUNTRY
1	7riwxSPUcQk	ON EST POSITIF AU CC	2020-08-11	UCpWaR3gNAQGsX	Tibo InShape	17	2020-08-12	1943013	165557	4198	15777	FR
2	AcBd_RH9JSw	PASSER UNE NUIT DAN	2020-08-11	UCUI7mwOyySfZzU	LeBouseuh	24	2020-08-12	1823553	346839	2767	8030	FR
3	JVm8P6kKgD0	FRANGLISH - My Salsa	2020-08-12	UCnwfc0O-LGEg8yf	Franglish	10	2020-08-12	196340	22362	187	1148	FR
4	JfgeD7xWy-w	L'ÉTÉ LE PLUS ECLATÉ	2020-08-11	UCMJ2VQ3-8zpyeAl	Fahd El	23	2020-08-12	311609	29561	418	3213	FR

12. Creation of table_youtube_category

	status
1	Table TABLE_YOUTUBE_FINAL successfully created.

13. Preview of table_youtube_category

	COUNTRY	CATEGORYID	CATEGORY_TITLE
1	FR	1	Film & Animation
2	FR	2	Autos & Vehicles
3	FR	10	Music
4	FR	15	Pets & Animals

14. Creation of table_youtube_final

	status
1	Table TABLE_YOUTUBE_FINAL successfully created.

15. Preview of table_youtube_final

	ID	VIDEO_ID	TITLE	PUBLISHEDAT	CHANNELID	CHANNELTITLE	CATEGORYID	CATEGORY_TITLE	TRENDING_DATE	VIEW_COUNT	LIKES	DISLIKE	COMMENT	COUNT
1	2861dd09-	79zF6Wh7rE	【オフショッ	2021-12-06	UCKnAJC2VKx	勝 友美-VICTORY C	24	Entertainment	2021-12-12	1058568	10883	1209	324	JP
2	1d5afd12-ε	EhJu6EhOWGε	[ETC]SEVEN	2021-12-07	UCpbTKp4B80	SEVENTEEN Japan	10	Music	2021-12-12	213225	28369	72	861	JP
3	9106d44b-	AxyzNFYdaJl	Q 夜18時~!	2021-12-06	UCAKXje2l7fV-	まっざき弁護士の労	27	Education	2021-12-12	505765	14512	413	378	JP
4	b4f552bf-e	zyYof9_oIro	[BE:FIRST]	2021-12-06	UCdhJWtQm6	ELLE Japan (エル	24	Entertainment	2021-12-12	424964	17446	184	817	JP